Peng Zhang

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EDUCATION

University of Wisconsin-Madison (UW-Madison) Madison, US
PhD in Electrical and Computer Engineering Sep 2025 - Present

- **GPA:** 4.0/4.0, **Advisors:** Prof. Manish Singh and Prof. Line Roald
- **Research:** Informativity for voltage monitoring and new load integration in distribution systems

University of Washington (UW) Seattle, US
MS in Electrical Engineering Sep 2023 - Jun 2025

- **GPA:** 3.95/4.0, **Advisor:** Prof. Baosen Zhang
- **Research:** Online optimal voltage control in power distribution networks

North China Electric Power University (NCEPU) Beijing, China
BEng in Electrical Engineering and Its Automation Sep 2019 - Jun 2023

- **GPA:** 93.08/100 (3.95/4.0), **Ranking in 3 years:** 1/55, 1/399, 7/396
- **Thesis:** High-Resolution Spatiotemporal Assessment of Onshore and Offshore Wind Energy and Long-Term System Planning, advised by Prof. Nian Liu

PUBLICATIONS

[1] **P. Zhang**, L. Roald and M. Singh. "Optimal Measurement Selection for Certifiable Voltage Monitoring in Power Distribution Systems," *arXiv preprint arXiv:2609.28785,* 2026. (Accepted to HICSS-60)

[2] **P. Zhang** and B. Zhang, "Optimal voltage control using online exponential barrier method," in *Proc. 57th North American Power Symposium (NAPS)*, 2025.

[3] **P. Zhang** and B. Zhang, "Online voltage regulation of distribution systems with disturbance-action controllers," *arXiv preprint arXiv:2412.00629*, 2024.

[4] Y. Zhang, L. Wang, **P. Zhang**, H. Luo, W. Shi, and X. Wang, "The nonlinear wave solutions and parameters discovery of the Lakshmanan-Porsezian-Daniel based on deep learning," *Chaos, Solitons & Fractals*, vol. 159, Art. no. 112155, 2022.

RESEARCH EXPERIENCE

Geometric and Algebraic Characterization of Informativity for Voltage Monitoring Madison, US
Research assistant with Prof. Manish Singh and Prof. Line Roald Sep 2025 - Present

- **Measurement selection:** Unveiled the complex geometry of the safety-guaranteed partial measurement set; formulated a bilevel problem to select the limited real-time measurements for network-wide voltage safety with the proposed monotone metric; derived a single-level reformulation via strong duality, and developed scenario reduction methods with computable optimality gaps; numerically validated computational efficiency and improved certification capability

  - Presented as a poster at PowerUp 2026 [Poster]

- **Geometric characterization:** Recast the optimization-based voltage safety assessment as a set-membership problem to reduce both the temporal and spatial complexity; analyzed the geometry of safeguarding and non-safeguarding measurement sets under the linearized power flow; proposed a dual solution-based algorithm to online construct the geometry through the supporting hyperplanes

  - Working paper: Online Voltage Safety Assessment via Geometric Characterization of Partial Measurement Set

Online Voltage Control in Distribution Systems under Model Inaccuracy Seattle, US
Research assistant with Prof. Baosen Zhang Jan 2024 - Aug 2025

- **Disturbance-action controller:** Developed an adaptive controller that exploits the temporal load correlations to coordinate active and reactive power outputs of inverters for voltage regulation; analytically derived the stability conditions and bounds on performance degradation under model inaccuracies; numerically demonstrated the robustness to communication delays and fluctuating load profiles and solar generation

- Presented as a poster at PESGM 2025 [Poster]

- **Online exponential barrier** Proposed an exponential barrier-based algorithm with real-time grid feedback to balance the control cost and voltage constraint satisfaction under model inaccuracies; derived the analytical safety barrier weights, the sufficient conditions for voltage safety at convergence, and the local exponential rate with an appropriate step size

  - Presented at NAPS 2025 [Slides]

Spatiotemporal Modeling and Planning of Large-scale Power Systems Beijing, China
Research assistant with Prof. Nian Liu Sep 2022 - Jun 2023

- **Resources assessment:** Evaluated the physical potential of PV and wind energy by processing hyperscale high-resolution numerical weather and geospatial data from Seniverse, NREL, NASA, etc., based on PVLIB and a modified wind output model; assessed economic potential by calculating the Levelized Cost of Electricity (LCOE)
- **System planning:** Formulated the large-scale linear/quadratic programming for the optimal system expansion and operation scheme within provincial grids with integration of utility-scale PV, wind, energy storage, etc.
- **Adequacy under extreme weather:** Characterized the nonlinear response of loads to the outdoor temperature; served as the team member to comprehensively evaluate the nationwide adequacy and flexibility of renewable energy during diverse extreme weather events, e.g., hot and cold waves. [Project Paper]

Interdisciplinary Research on Machine Learning and Nonlinear Dynamics Beijing, China
Research assistant with Prof. Lei Wang May 2020 - Jun 2023

- **Analytical methods:** Applied Linear Stability Analysis (LSA), Darboux Transformation (DT) to explore the evolution of solution dynamics for governing high-order and high-dimensional nonlinear PDEs
- **Data-driven methods:** Leveraged Physics-Informed Neural Network (PINN) to both discover the mechanism of control equations (backwards) and infer the solutions (forwards) under corresponding conditions; extended PINN to (2+1)-d in the context of DS I equation that describes the wave dynamics in shallow water; predicted the evolution of rogue waves and explored the transition conditions among three families of lump waves

INTERNSHIP

GE Vernova, Bothell, WA Jun 2024 - Aug 2024
Software Engineer Intern – Power Systems

- Gained hands-on experience with GE Vernova's software *Platform and Habitat* delivered to U.S. utilities for analysis and operation of transmission systems
- Developed an optimal scheduling framework for controllable reactive power resources, including automatic voltage regulators (AVRs) and shunt capacitor banks, to minimize transmission system losses; and implemented the framework as a Python module for GE Vernova's Energy Management System (EMS) software
- Solved the cost-effective actions to satisfy utility-specific voltage requirements and applied chance-constrained optimization to improve system robustness against large uncertainties from renewable energy resources

AWARDS & HONORS

Novotny Welcome Awards, UW-Madison Oct 2025
NAPS Student Travel Award, Hartford CT Oct 2025
IEEE PES GM Student program, Austin TX Jul 2025
Outstanding Graduated Undergrad in Beijing (4%), NCEPU Jun 2023
Extra-high Voltage (EHV) Scholarship (0.4%), NCEPU Sep 2022
First-class Schneider Electric Scholarship (1%), NCEPU Sep 2021
First-class Academic Scholarship (5%), NCEPU Sep 2020, 2021, 2022

SKILLS

Programming: Python, MATLAB, Mathematica, C, Julia
Tools: (*Optimization*) Gurobi/Mosek/CPLEX, CVX; (*Power Systems*) MATPOWER/Pandapower/PyPSA, Multisim, PVLIB, PowerWorld, COMSOL; (*Machine Learning*) PyTorch and TensorFlow

COMPETITION

Mathematical Contest in Modeling (MCM) Feb 2021
Team Leader; ***Finalist Winners, Top 2% Worldwide***

- Formulated a framework of wildfire risk assessment and resource allocation using NASA climate data.
- Adapted the ID3 decision tree to classify wildfire risk across discretized geographic regions, applied time-series analysis (TSA) to forecast fire occurrence, and derived a data-driven drone procurement strategy for rapid wildfire response in Australia.

EXTRACURRICULAR ACTIVITIES

Volunteer for Discovery Day 2025, UW May 2024

- Guided 4th–8th grade students and their parents in hands-on science exhibition activities and promoted engagement within the interdisciplinary engineering community.

Captain of Soccer Team, Department of Mathematics and Physics at NCEPU Sep 2020 - Jun 2021

- Took charge of the annual recruitment, weekly training, match deployment, etc.